MECH 325 Machine Elements

Tutorial 3 – Flexible Drives

1. Adapted from **Shigley, Problem 17-29 (all editions)**.

   A mine hoist uses a 2-inch 6 × 19 monitor-steel wire rope. The rope is used to haul maximum loads of 4 tons from a 480 ft. shaft. The drum has a diameter of 6 feet and the sheaves are of good-quality cast steel, and the smallest is 3 feet in diameter.

   Part 1: To be done by the student

   a. Using a maximum hoisting speed of 1200 feet/minute and a maximum acceleration of 2 ft/s^2, estimate the stresses in the wire rope.

   b. Estimate the various factors of safety

2. Adapted from **Shigley, Problem 17-4**. (If time allows)

   A 6-in-wide polyamide A-3 flat belt is used to connect two large 48″ cast-iron pulleys spaced 16 feet apart. The belt is to transmit 60HP at a pulley speed of 380 rpm. The service factor K_s of 1.10 and the design factor is 1.0.

   Part 1: To be done by the students

   a. Find F_c, F_i, F_{1a}, and F_2

   Part 2: To be done by the TA.

   b. What is the maximum allowable power H_a
   c. What is the effective friction, f''? Will the belt slip?
   d. Find the belt length
   e. Can we reduce the pre-tension to improve belt life? If so, determine the new values of tension.

MECH 325 Tutorial Additional Information

Tutorial 3: Flexible Drives

Table 17–2

Properties of Some Flat- and Round-Belt Materials. (Diameter = d, thickness = t, width = w)

Material	Specification	Size, in	Minimum Pulley Diameter, in	Allowable Tension per Unit Width at 600 ft/min, lbf/in	Specific Weight, lbf/in^3	Coefficient of Friction
Leather	1 ply	$t = \frac{11}{64}$	3	30	0.035–0.045	0.4
		$t = \frac{13}{64}$	$3\frac{1}{2}$	33	0.035–0.045	0.4
	2 ply	$t = \frac{18}{64}$	$4\frac{1}{2}$	41	0.035–0.045	0.4
		$t = \frac{20}{64}$	6[a]	50	0.035–0.045	0.4
		$t = \frac{23}{64}$	9[a]	60	0.035–0.045	0.4
Polyamide[b]	F–0[c]	$t = 0.03$	0.60	10	0.035	0.5
	F–1[c]	$t = 0.05$	1.0	35	0.035	0.5
	F–2[c]	$t = 0.07$	2.4	60	0.051	0.5
	A–2[c]	$t = 0.11$	2.4	60	0.037	0.8
	A–3[c]	$t = 0.13$	4.3	100	0.042	0.8
	A–4[c]	$t = 0.20$	9.5	175	0.039	0.8
	A–5[c]	$t = 0.25$	13.5	275	0.039	0.8
Urethane[d]	$w = 0.50$	$t = 0.062$	See	5.2[e]	0.038–0.045	0.7
	$w = 0.75$	$t = 0.078$	Table	9.8[e]	0.038–0.045	0.7
	$w = 1.25$	$t = 0.090$	17–3	18.9[e]	0.038–0.045	0.7
	Round	$d = \frac{1}{4}$	See	8.3[e]	0.038–0.045	0.7
		$d = \frac{3}{8}$	Table	18.6[e]	0.038–0.045	0.7
		$d = \frac{1}{2}$	17–3	33.0[e]	0.038–0.045	0.7
		$d = \frac{3}{4}$		74.3[e]	0.038–0.045	0.7

[a] Add 2 in to pulley size for belts 8 in wide or more.

[b] *Source: Habasit Engineering Manual*, Habasit Belting, Inc., Chamblee (Atlanta), Ga.

[c] Friction cover of acrylonitrile-butadiene rubber on both sides.

[d] *Source:* Eagle Belting Co., Des Plaines, Ill.

[e] At 6% elongation; 12% is maximum allowable value.

Table 17–4

Pulley Correction Factor C_p for Flat Belts*

Material	Small-Pulley Diameter, in					
	1.6 to 4	4.5 to 8	9 to 12.5	14, 16	18 to 31.5	Over 31.5
Leather	0.5	0.6	0.7	0.8	0.9	1.0
Polyamide, F–0	0.95	1.0	1.0	1.0	1.0	1.0
F–1	0.70	0.92	0.95	1.0	1.0	1.0
F–2	0.73	0.86	0.96	1.0	1.0	1.0
A–2	0.73	0.86	0.96	1.0	1.0	1.0
A–3	—	0.70	0.87	0.94	0.96	1.0
A–4	—	—	0.71	0.80	0.85	0.92
A–5	—	—	—	0.72	0.77	0.91

*Average values of C_p for the given ranges were approximated from curves in the *Habasit Engineering Manual*, Habasit Belting, Inc., Chamblee (Atlanta), Ga.

Table 17–24

Wire-Rope Data *Source:* Compiled from *American Steel and Wire Company Handbook.*

Rope	Weight per meter, $(10^{-3})N$	Minimum Sheave Diameter, mm	Standard Sizes d, mm	Material	Size of Outer Wires	Modulus of Elasticity,* GPa	Strength,† MPa
6 × 7 haulage	$33.92d^2$	$42d$	6 – 38	Monitor steel	$d/9$	96	690
				Plow steel	$d/9$	96	608
				Mild plow steel	$d/9$	96	524
6 × 19 standard hoisting	$36.18d^2$	$26d$–$34d$	6 – 70	Monitor steel	$d/13$–$d/16$	83	730
				Plow steel	$d/13$–$d/16$	83	640
				Mild plow steel	$d/13$–$d/16$	83	550
6 × 37 special flexible	$35.05d^2$	$18d$	6 – 90	Monitor steel	$d/22$	76	690
				Plow steel	$d/22$	76	608
8 × 19 extra flexible	$32.79d^2$	$21d$–$26d$	6 – 38	Monitor steel	$d/15$–$d/19$	69	634
				Plow steel	$d/15$–$d/19$	69	550
7 × 7 aircraft	$38.45d^2$	—	1.6 – 10	Corrosion-resistant steel	—	—	850
				Carbon steel	—	—	850
7 × 9 aircraft	$39.58d^2$	—	3 – 36	Corrosion-resistant steel	—	—	930
				Carbon steel	—	—	986
19-wire aircraft	$48.62d^2$	—	0.8 – 8	Corrosion-resistant steel	—	—	1137
				Carbon steel	—	—	1137

*The modulus of elasticity is only approximate; it is affected by the loads on the rope and, in general, increases with the life of the rope.

†The strength is based on the nominal area of the rope. The figures given are only approximate and are based on 25-mm rope sizes and 6-mm aircraft-cable sizes.

Rope	Weight per Foot, lbf	Minimum Sheave Diameter, in	Standard Sizes d, in	Material	Size of Outer Wires	Modulus of Elasticity,* Mpsi	Strength,† kpsi
6 × 7 haulage	$1.50d^2$	$42d$	$\frac{1}{4}$ to $1\frac{1}{2}$	Monitor steel	$d/9$	14	100
				Plow steel	$d/9$	14	88
				Mild plow steel	$d/9$	14	76
6 × 19 standard hoisting	$1.60d^2$	$26d$ to $34d$	$\frac{1}{4}$ to $2\frac{3}{4}$	Monitor steel	$d/13$ to $d/16$	12	106
				Plow steel	$d/13$ to $d/16$	12	93
				Mild plow steel	$d/13$ to $d/16$	12	80
6 × 37 special flexible	$1.55d^2$	$18d$	$\frac{1}{4}$ to $3\frac{1}{2}$	Monitor steel	$d/22$	11	100
				Plow steel	$d/22$	11	88
8 × 19 extra flexible	$1.45d^2$	$21d$ to $26d$	$\frac{1}{4}$ to $1\frac{1}{2}$	Monitor steel	$d/15$ to $d/19$	10	92
				Plow steel	$d/15$ to $d/19$	10	80
7 × 7 aircraft	$1.70d^2$	—	$\frac{1}{16}$ to $\frac{3}{8}$	Corrosion-resistant steel	—	—	124
				Carbon steel	—	—	124
7 × 9 aircraft	$1.75d^2$	—	$\frac{1}{8}$ to $1\frac{3}{8}$	Corrosion-resistant steel	—	—	135
				Carbon steel	—	—	143
19-wire aircraft	$2.15d^2$	—	$\frac{1}{32}$ to $\frac{5}{16}$	Corrosion-resistant steel	—	—	165
				Carbon steel	—	—	165

*The modulus of elasticity is only approximate; it is affected by the loads on the rope and, in general, increases with the life of the rope.

†The strength is based on the nominal area of the rope. The figures given are only approximate and are based on 1-in rope sizes and $\frac{1}{4}$-in aircraft-cable sizes.

Source: Compiled from *American Steel and Wire Company Handbook.*

Table 17–25

Minimum Factors of Safety for Wire Rope*

Source: Compiled from a variety of sources, including ANSI A17.1-1978.

Track cables	3.2
Guys	3.5
Mine shafts, m:	
Up to 152.5	8.0
305–610	7.0
610–915	6.0
Over 915	5.0
Hoisting	5.0
Haulage	6.0
Cranes and derricks	6.0
Electric hoists	7.0
Hand elevators	5.0
Private elevators	7.5
Hand dumbwaiter	4.5
Grain elevators	7.5
Passenger elevators, m/s:	
0.25	7.60
1.52	9.20
4.06	11.25
6.10	11.80
7.62	11.90
Freight elevators, m/s:	
0.25	6.65
1.52	8.20
4.06	10.00
6.10	10.50
7.62	10.55
Powered dumbwaiters, m/s:	
0.25	4.8
1.52	6.6
4.06	8.0

*Use of these factors does not preclude a fatigue failure.

Table 17–26

Maximum Allowable Bearing Pressures of Ropes on Sheaves (in MPa)

Source: Wire Rope Users Manual, AISI, 1979.

	Sheave Material				
Rope	Wood[a]	Cast Iron[b]	Cast Steel[c]	Chilled Cast Irons[d]	Manganese Steel[e]
Regular lay:					
6 × 7	1.0	2.1	3.8	4.5	10.1
6 × 19	1.7	3.3	6.2	7.6	16.6
6 × 37	2.1	4.0	7.4	9.1	20.7
8 × 19	2.4	4.7	8.7	10.7	24.1
Lang lay:					
6 × 7	1.1	2.4	4.1	4.9	11.4
6 × 19	1.9	3.8	6.9	8.3	19.0
6 × 37	2.3	4.6	8.1	10.0	22.8

[a]On end grain of beech, hickory, or gum.

[b]For H_B (min.) = 125.

[c]30–40 carbon; H_B (min.) = 160.

[d]Use only with uniform surface hardness.

[e]For high speeds with balanced sheaves having ground surfaces.

Table 17-26 Maximum Allowable Bearing Pressure of Ropes on Sheaves (in psi)

	Sheave Material				
Rope	Wood[a]	Cast Iron[b]	Cast Steel[c]	Chilled Cast Irons[d]	Manganese Steel[e]
Regular lay:					
6 × 7	150	300	550	650	1470
6 × 19	250	480	900	1100	2400
6 × 37	300	585	1075	1325	3000
8 × 19	350	680	1260	1550	3500
Lang lay:					
6 × 7	165	350	600	715	1650
6 × 19	275	550	1000	1210	2750
6 × 37	330	660	1180	1450	3300

[a]On end grain of beech, hickory, or gum.

[b]For H_B (min.) = 125.

[c]30–40 carbon; H_B (min.) = 160.

[d]Use only with uniform surface hardness.

[e]For high speeds with balanced sheaves having ground surfaces.

Table 17–27

Some Useful Properties of 6 × 7, 6 × 19, and 6 × 37 Wire Ropes

Wire Rope	Weight per meter w, (10^{-3}) N/m	Weight per meter Including Core w, (10^{-3}) N/m	Minimum Sheave Diameter D, mm	Better Sheave Diameter D, mm	Diameter of Wires d_w, mm	Area of Metal A_m, mm^2	Rope Young's Modulus E_r, GPa
6 × 7	$33.92d^2$		$42d$	$72d$	$0.111d$	$0.38d^2$	13×10^6
6 × 19	$36.18d^2$	$39.8d^2$	$30d$	$45d$	$0.067d$	$0.40d^2$	12×10^6
6 × 37	$35.05d^2$	$38.67d^2$	$18d$	$27d$	$0.048d$	$0.40d^2$	12×10^6

Table 17–27

Some Useful Properties of 6 × 7, 6 × 19, and 6 × 37 Wire Ropes

Wire Rope	Weight per Foot w, lbf/ft	Weight per Foot Including Core w, lbf/ft	Minimum Sheave Diameter D, in	Better Sheave Diameter D, in	Diameter of Wires d_w, in	Area of Metal A_m, in^2	Rope Young's Modulus E_r, psi
6 × 7	$1.50d^2$		$42d$	$72d$	$0.111d$	$0.38d^2$	13×10^6
6 × 19	$1.60d^2$	$1.76d^2$	$30d$	$45d$	$0.067d$	$0.40d^2$	12×10^6
6 × 37	$1.55d^2$	$1.71d^2$	$18d$	$27d$	$0.048d$	$0.40d^2$	12×10^6

Figure 17–21

Experimentally determined relation between the fatigue life of wire rope and the sheave pressure.

